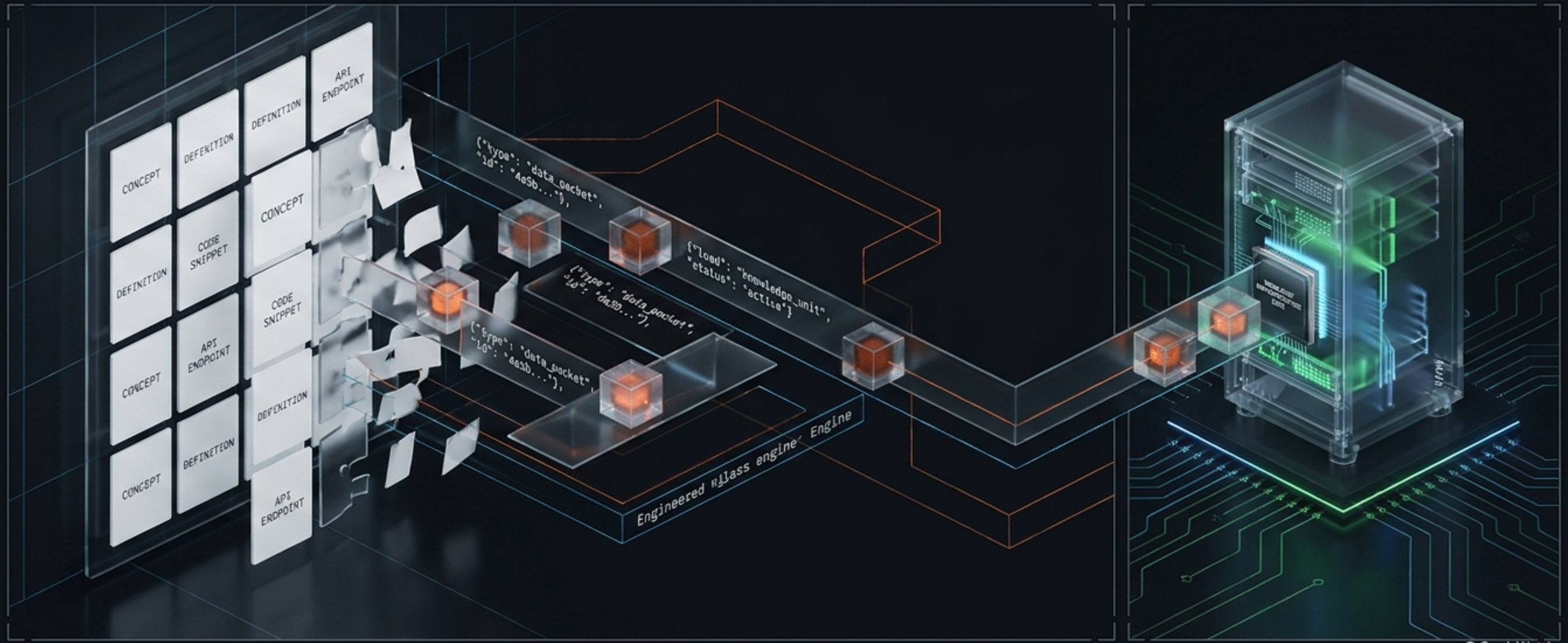
AnkiTov Community Tour

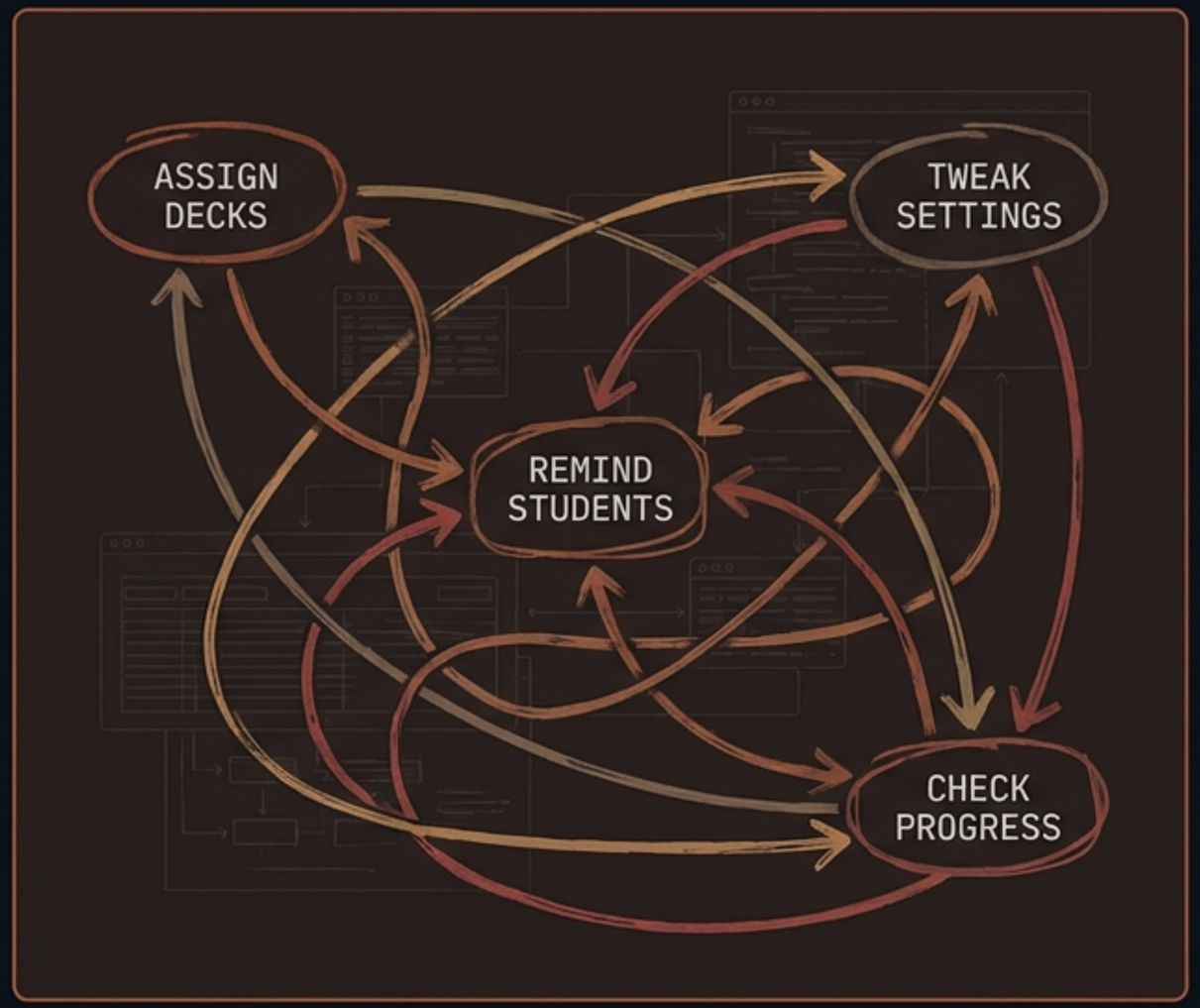
A Developer's Guide to Zero-Intervention Knowledge Infrastructure

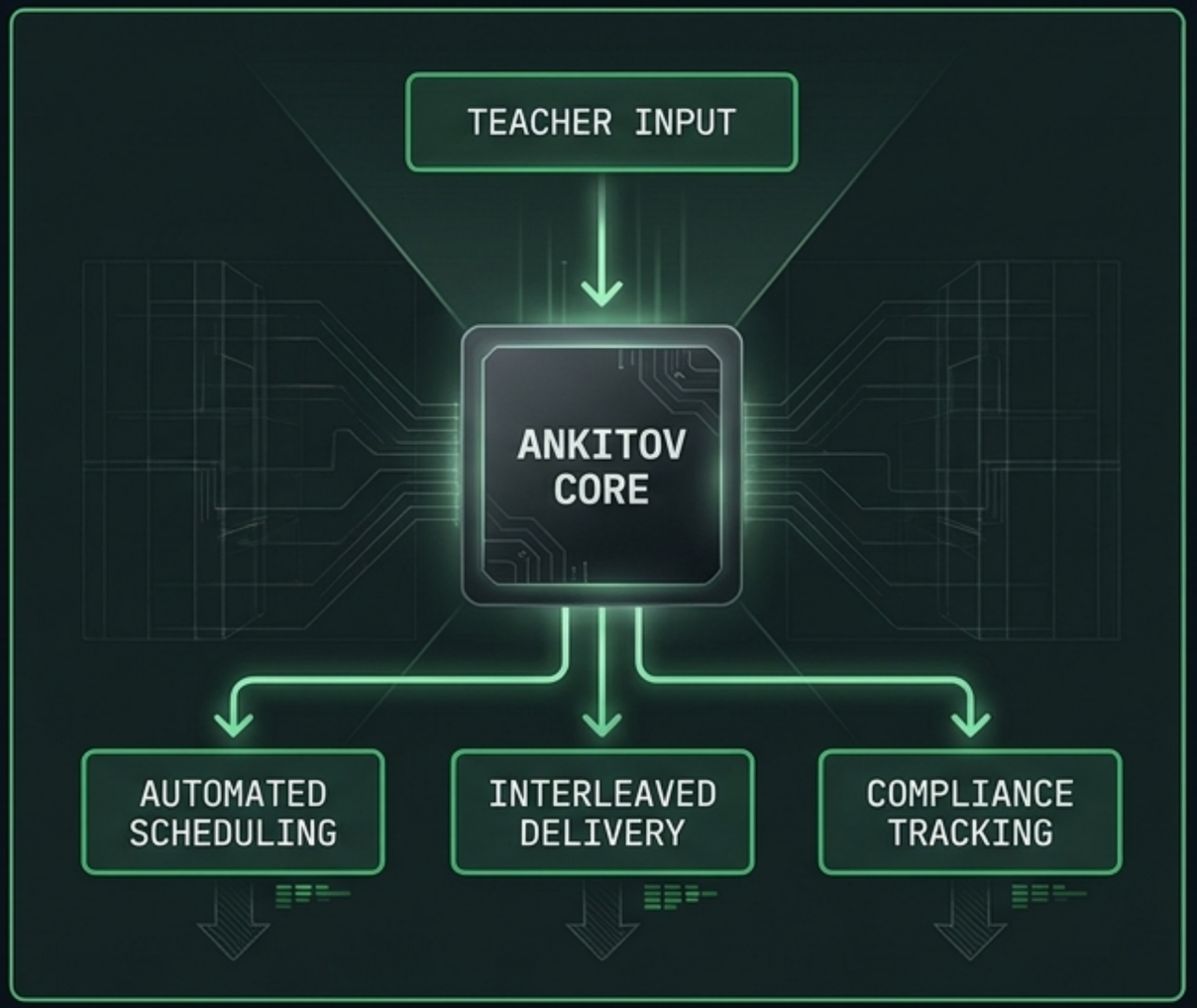


Gemini Notebook

THE FRICTION OF INSTITUTIONAL SPACED REPETITION, SOLVED.

The teacher assigns once; the system schedules, delivers, interleaves, and reports.



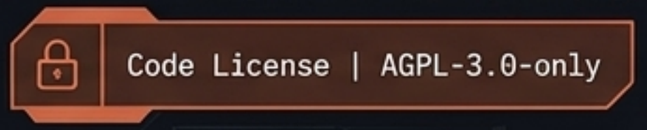


"THE TEACHER ASSIGNS ONCE; THE SYSTEM SCHEDULES, DELIVERS, INTERLEAVES, AND REPORTS."

Gemini Notebook

WRAPPING A LEGACY ENGINE IN CLOUD-NATIVE INFRASTRUCTURE.

	STOCK ANKI	ANKITOV
USER SCOPE	Individual	Institutional
SCHEDULING AUTHORITY	Client-side logic	Server-side FSRS-6 logic
ADMINISTRATION	Solo management	Class-wide compliance tracking
TECH STACK	Python / Qt / SQLite	Rust 2021 + Loco.rs + Axum + SeaORM





Gemini Notebook

THE FIVE-LAYER ISOLATION ARCHITECTURE

CLOUD GATEWAY
Routing, authentication, tenancy

STOCK ANKI CORE
Unmodified scheduling and review behavior

CUSTODY ADD-ON
Server-side enforcement, no client DRM

HTML5 STREAM
Browser streaming via Selkies

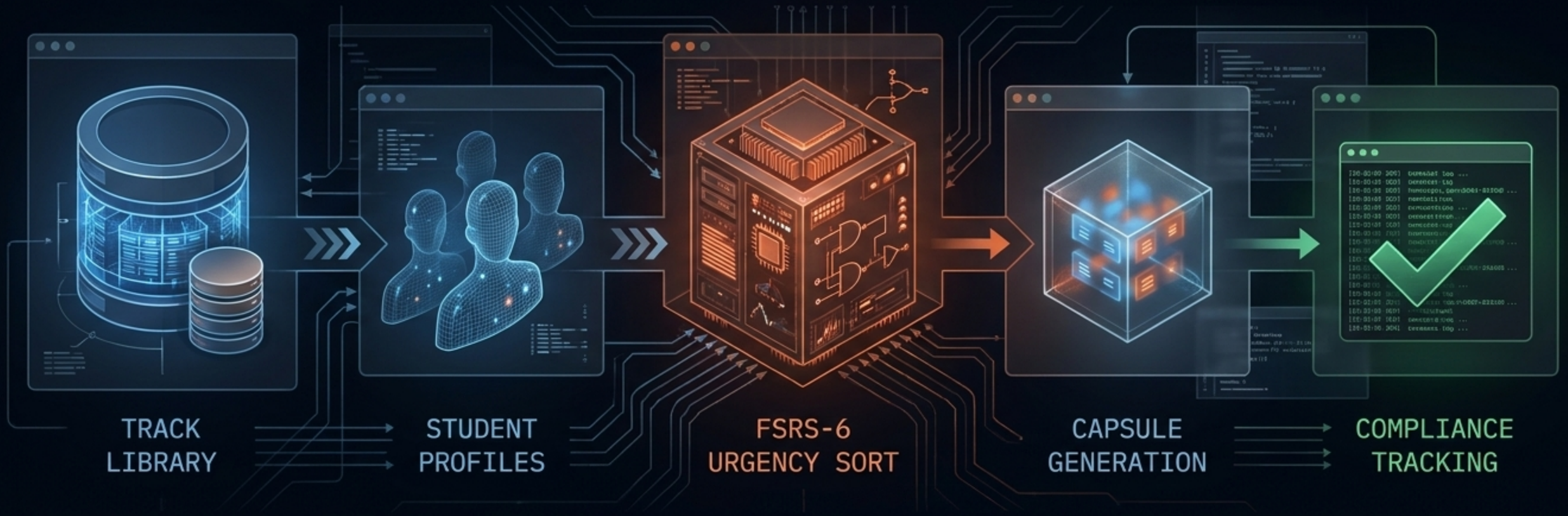
CLIENT
Any browser, or native sync client



Upstream Anki behavior stays entirely stock.

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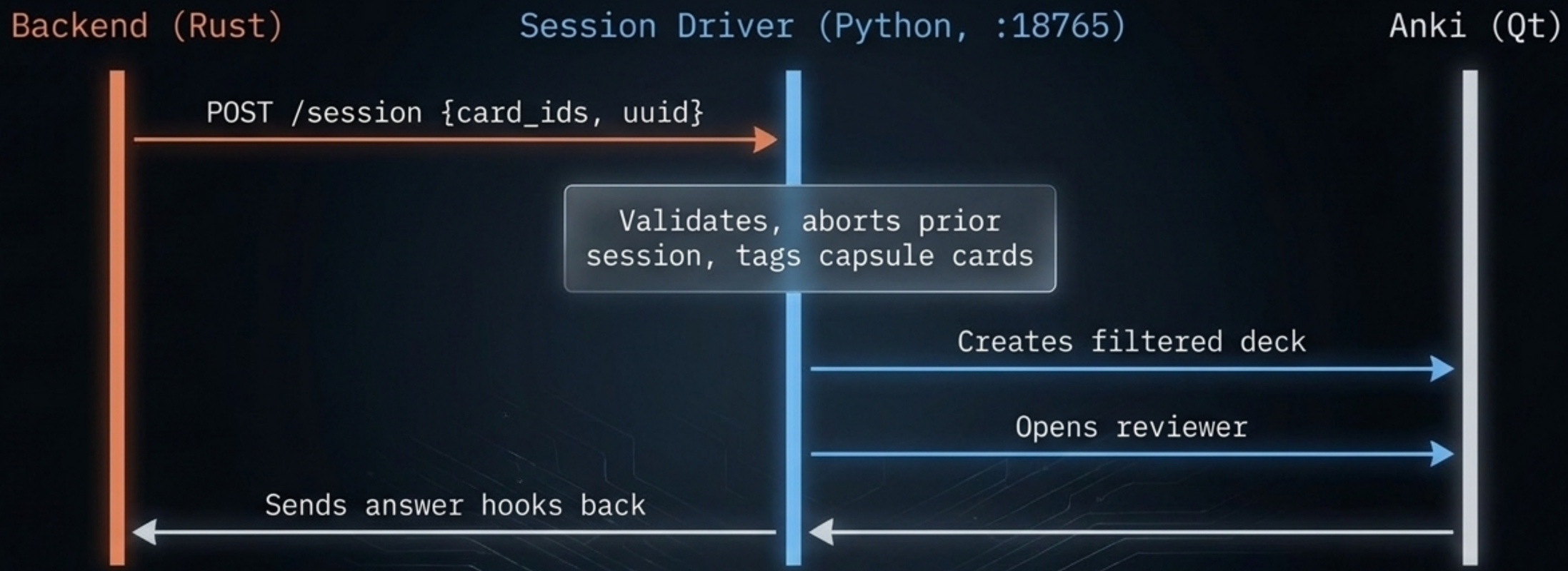
THE INTERLEAVED MASTERY PIPELINE (IMP)



Capsule: A pre-computed remediation set of card IDs, computed entirely server-side.

Gemini Notebook

CAPSULE DELIVERY VIA THE SESSION DRIVER ADD-ON



⚠ AnkiConnect (:8765) alone fails here. GUI widget switches on background threads silently fail.

Solution: Command-file handshake polled by a main-thread QTimer.

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Zero-install browser streaming with Selkies

WebSockets & Wayland/X11 Protocol

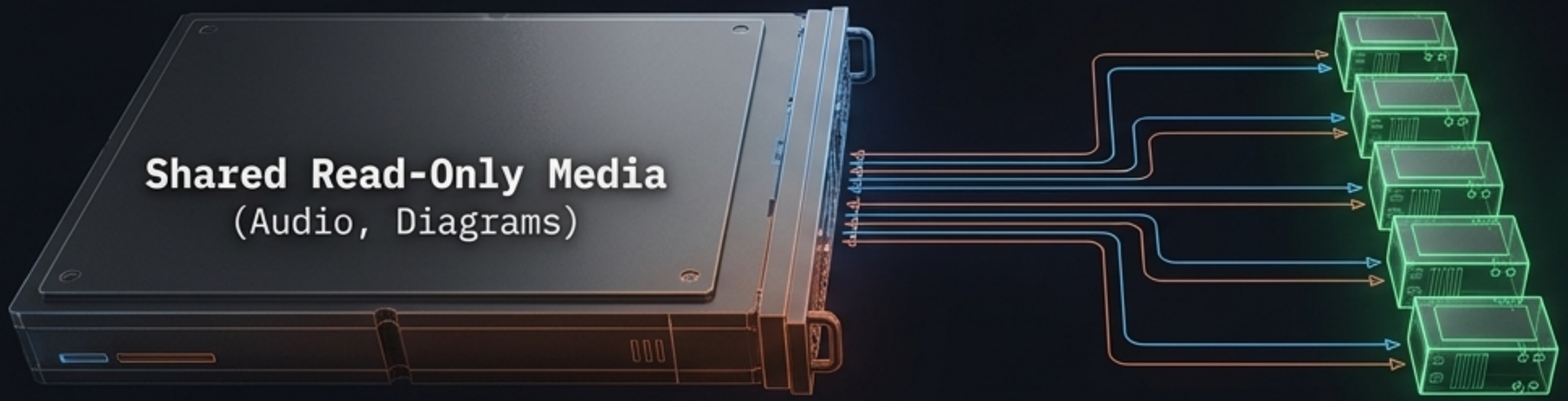
1

150-250 kbps
requirement

2

+50-80 ms frame latency
(Under 1% of human read-think cycle)

Scale Engineering



Warm Containers
(<1s attach time)

~250–350 MB RAM per container footprint. ~95% disk/RAM saved per instance.

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CONTINUING ON A PERSONAL DEVICE WITHOUT LOCK-IN

THE IDENTITY MAP

AnkiTov Account (ID, Email)

Container Profile
e.g., stu_<id>

Self-hosted Anki
Sync Server (:8080)



SYNC_USER1="stu_<id>:<pass>"



Desktop



AnkiDroid

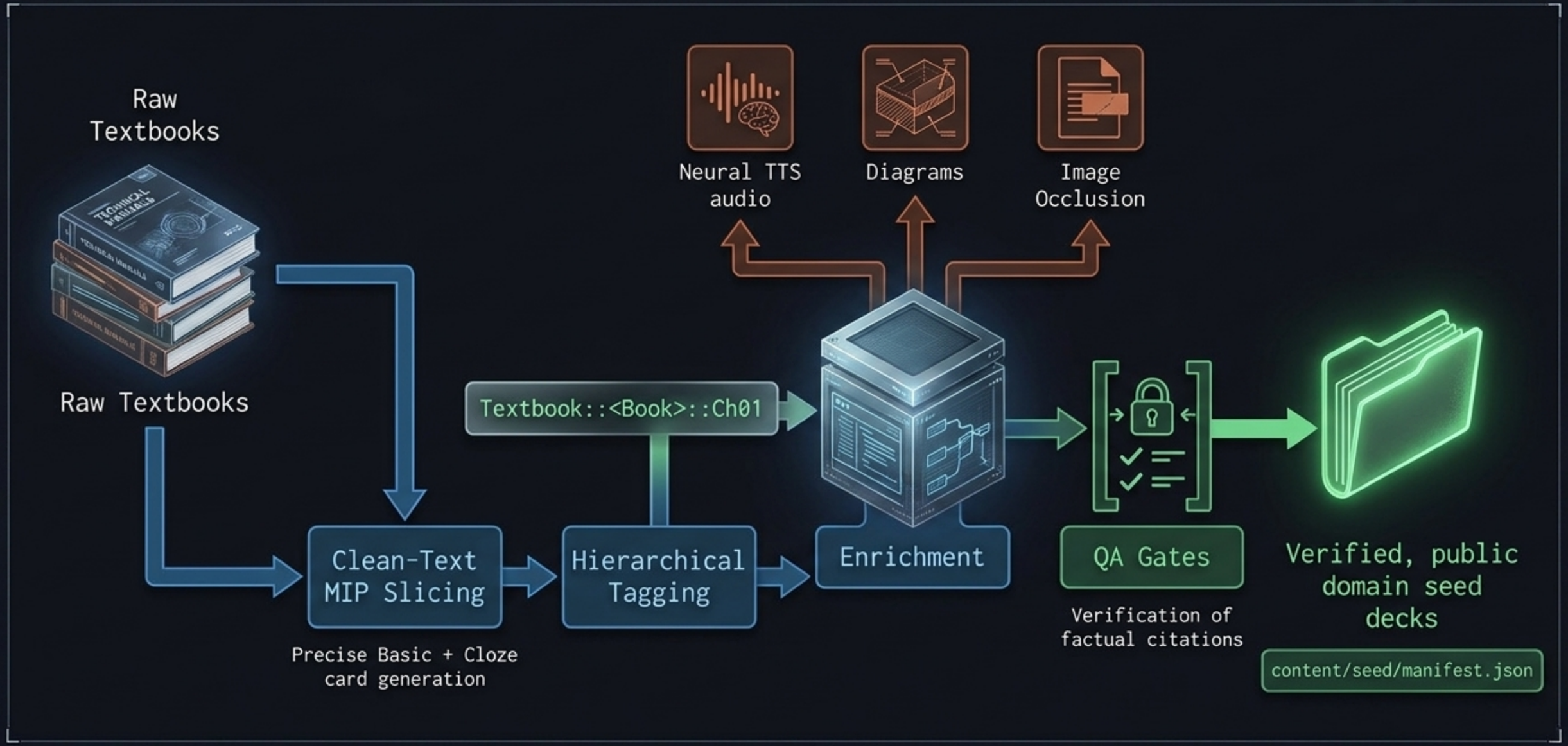


AnkiMobile

Uses Anki’s native multi-user sync and last-write-wins conflict resolution.
No re-downloads. No export/import required.

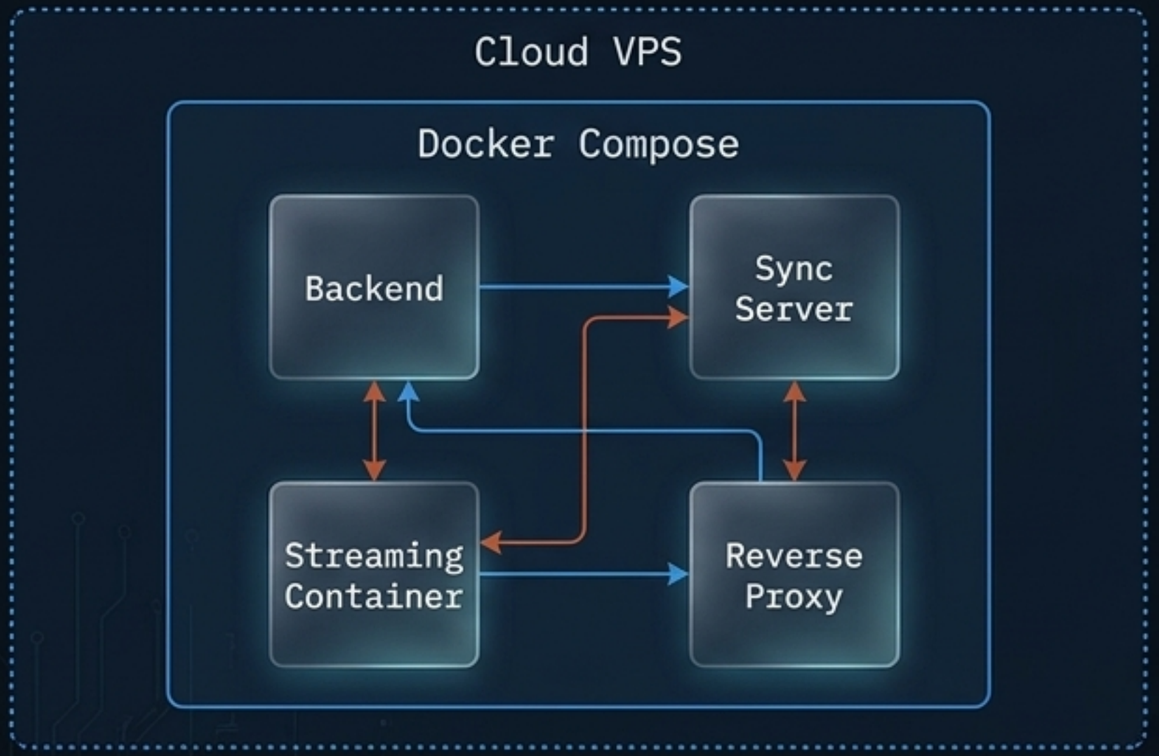
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The Content Engine and the Minimum Information Principle



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Ops, Deployment, and the Server Math



Capacity Mathematics:

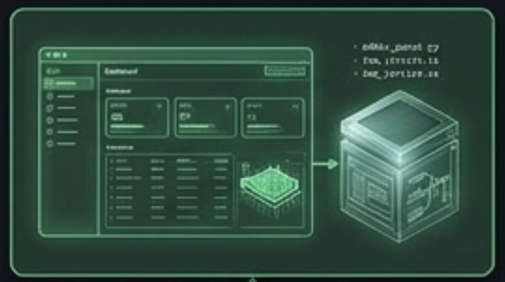
300–350 peak concurrent sessions / **10,000** daily learners

Fits effortlessly on **1–2** bare-metal servers.

```
bash scripts/bootstrap.sh
cd backend && cargo run start
# Dashboard: :5150/dashboard | OpenAPI: :5150/scalar
```

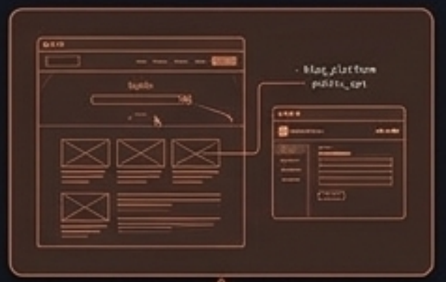
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The 7-Prong Project Roadmap









Management Console + IMP

Web Deployment Infrastructure

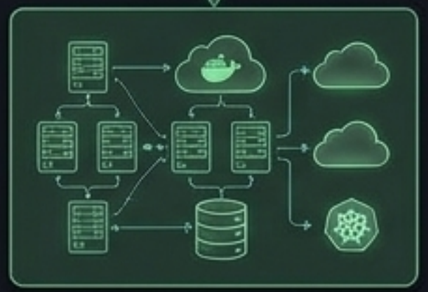
Class-Wide Practice Sessions

Gaming Console Streaming

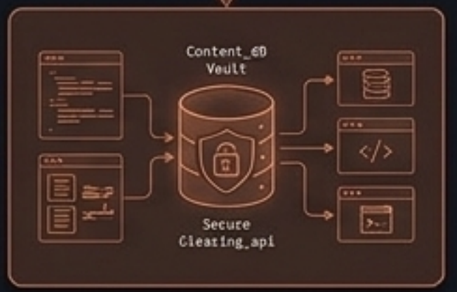
Commercial Analytics

Content Clearing House

Public Platform & Blog





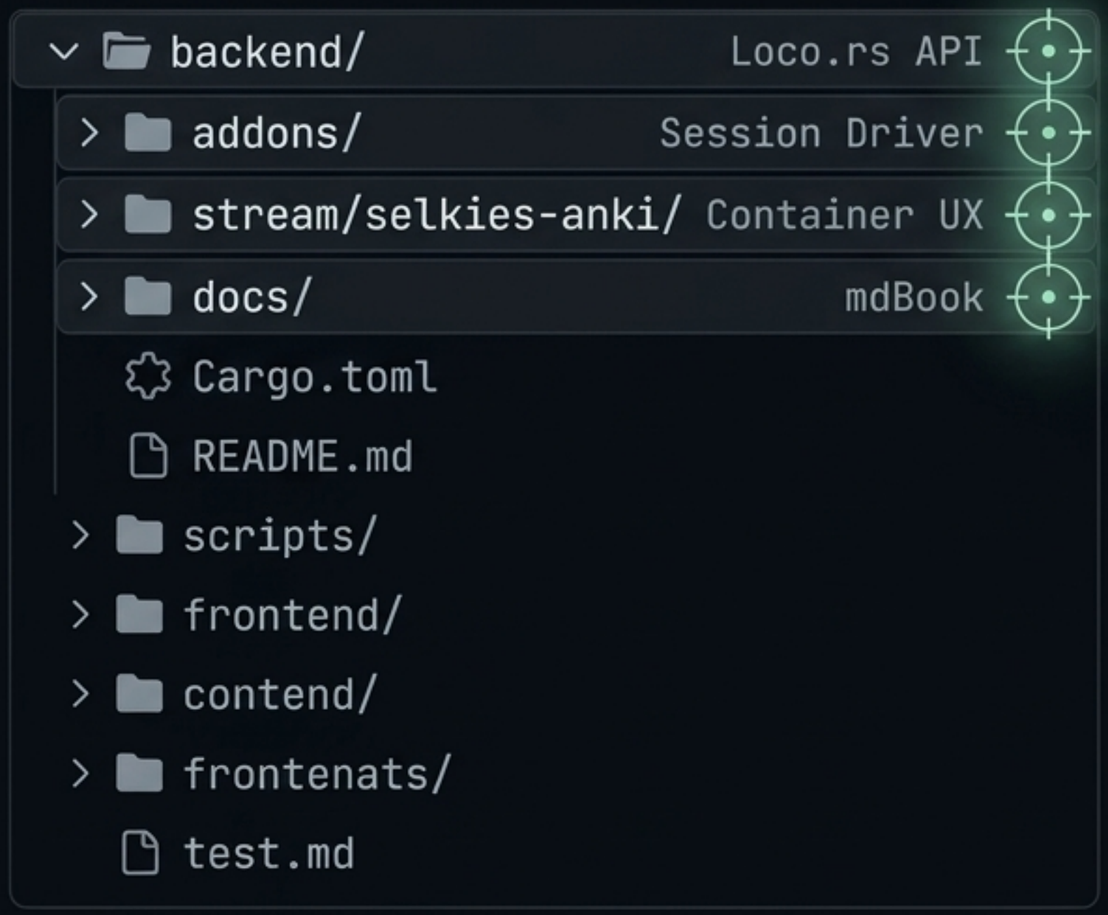




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Join the build: Surfaces, standards, and version control

Map of the Repo



Contribution Conventions

1. All commits use Jujutsu (jj).
2. Zero unsafe Rust.
3. No new unwrap()/expect() in production paths.
4. Test backend via cargo nextest run.
5. E2E flows cover Signup → Session → Sync.

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